Long term simulations

Perennial crops and crop rotations

Marie Launay
(INRAE, US AgroClim, Avignon, France)

Why a specific topic ?

- Most of the model modifications leading to the Stics version 10.0.0 were done to improve the model ability to simulate perennial crops and crop rotations on the long-term.

  - ➢ Perennial crops have specificities compared to annual crops :
  - ✓ Winter dormancy and restart of growth in spring
  - ✓ Remobilisation of C & N reserves from one year to another
  - ✓ Remobilisation of C & N reserves after cutting
  - ✓ A more or less developped root system which turn-over plays a major role in C&N cycles (mineral N availability and C and N storage in soil)

- Chaining simulations is not as easy at it mays appear…

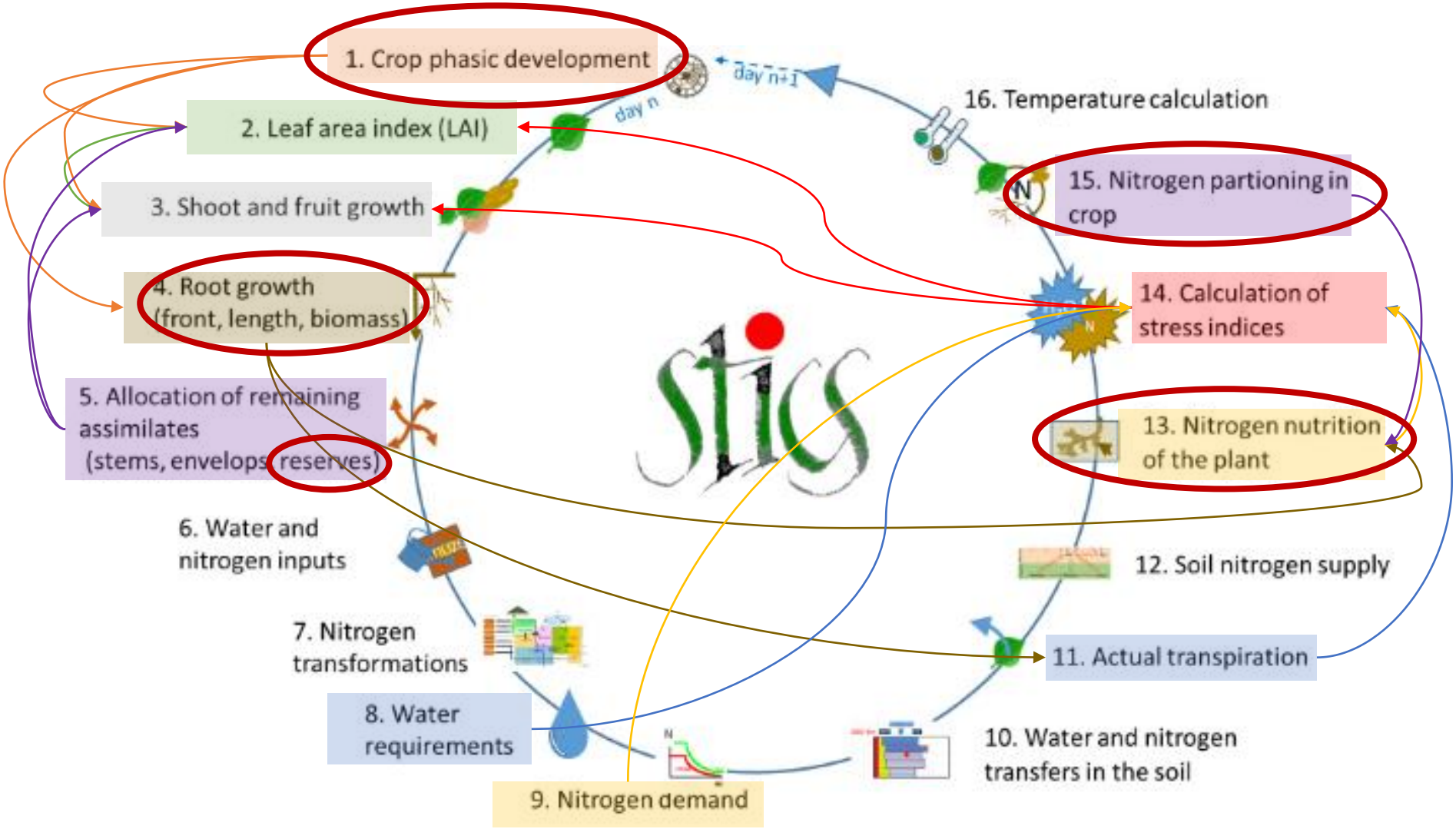
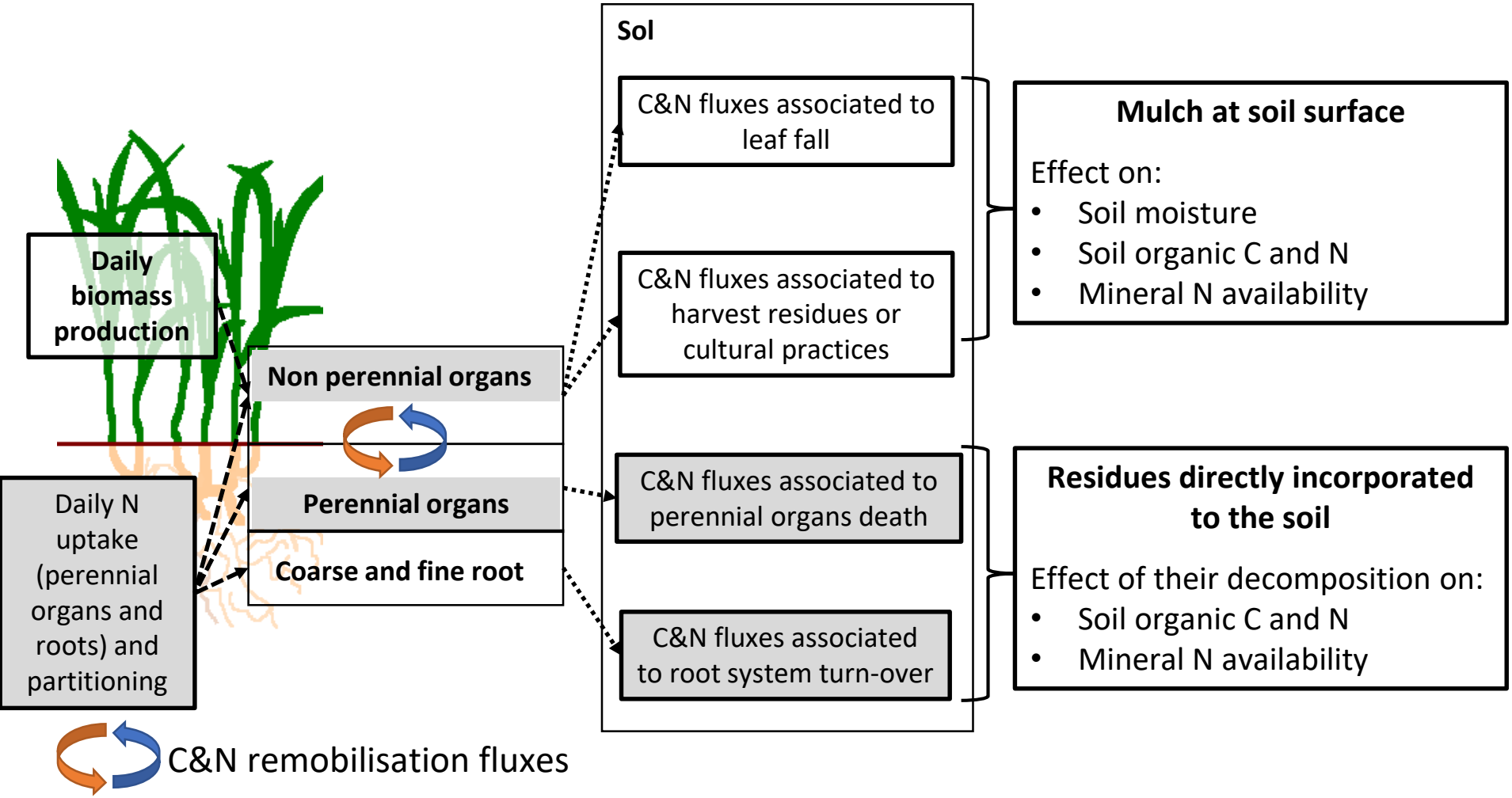


Figure 2.4. Simplified diagram of the order in STICS between processes within the daily loop.


Sol
C&N fluxes associated to leaf fall
C&N fluxes associated to harvest residues or cultural practices
C&N fluxes associated to perennial organs death
C&N fluxes associated to root system turn-over
Mulch at soil surface
Effect on:
• Soil moisture
• Soil organic C and N
• Mineral N availability
Residues directly incorporated to the soil
Effect of their decomposition on:
• Soil organic C and N
• Mineral N availability
Daily biomass production
Non perennial organs
Perennial organs
Coarse and fine root
Daily N uptake (perennial organs and roots) and partitioning
C&N remobilisation fluxes

Perennial crops

Development, root system and C&N cycling

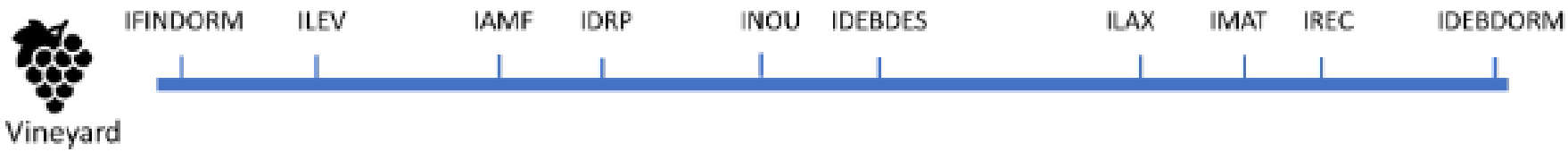
Crop development (Chapter 3 pp 47-65)

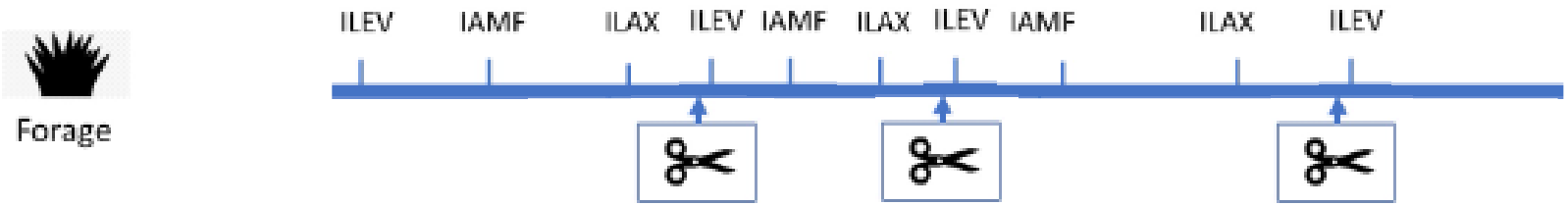
Vegetative stages (Leaf Area Index)	**Reproductive stages (harvested organs filling)**
IPLT: seedling	
IGER: germination	
IDEBDORM et IFINDORM: beginning and end of dormancy	
ILEV: Emergence for annual crops or Budbreak for perennial crops	
ILET: End of plantlet sensitivity to frost	IFLO: flowering
IAMF: maximum acceleration of leaf growth, end of juvenile phase	IDRP: Onset of filling of harvested organs
ILAX: End of LAI growth, emission of new leaves stop	INOU: end of setting (indeterminate crops)
	IDEBDES: onset of water dynamics in fruits
	IMAT: physiological maturity
IREC: harvest	

Crop development (Chapter 3 pp 47-65)

- Specificities of perennial crops :
  - Perennial crops can be sown: in this case the first development phases are those of annual crops (sowing, germination and emergence)

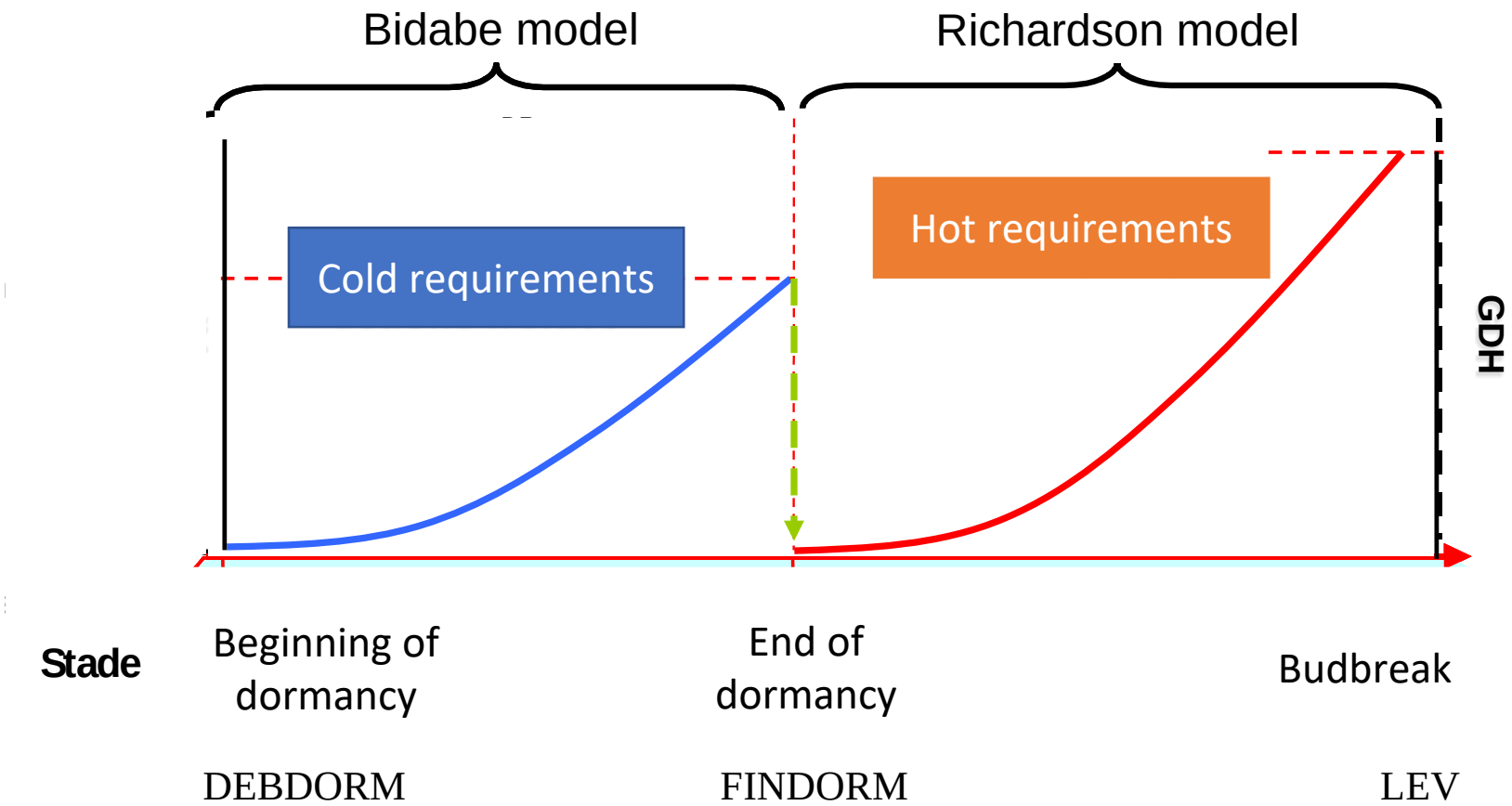
  - Perennial crops can be planted (with a rhizome for example) or they can be in place (such as vineyards) :





Crop development (Chapter 3 pp 47-65)

- End of dormancy and bud-break for perennial crops

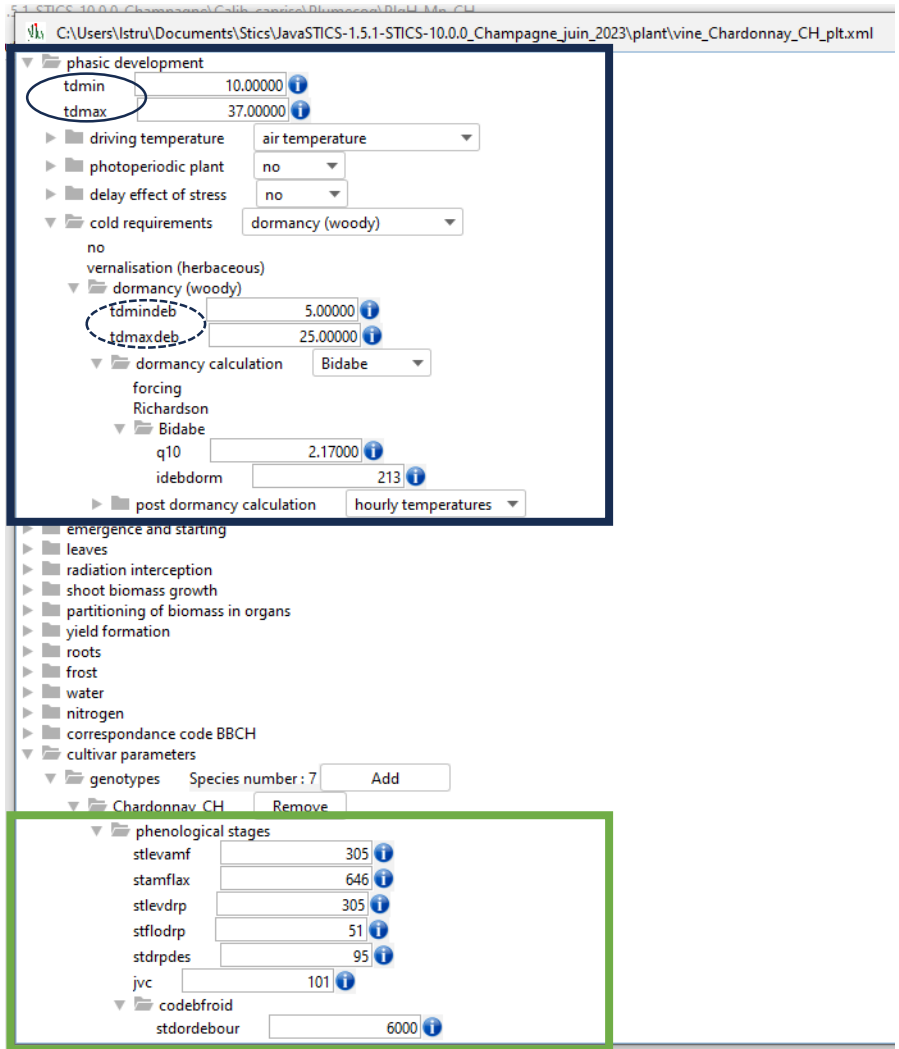


Garcia de Cortazar-Atauri et al., 2009

NEW

Crop development

- Plant parameters and options for the simulation of development are available in the plant file:



Root system (Chapter 5 pp 93-117)

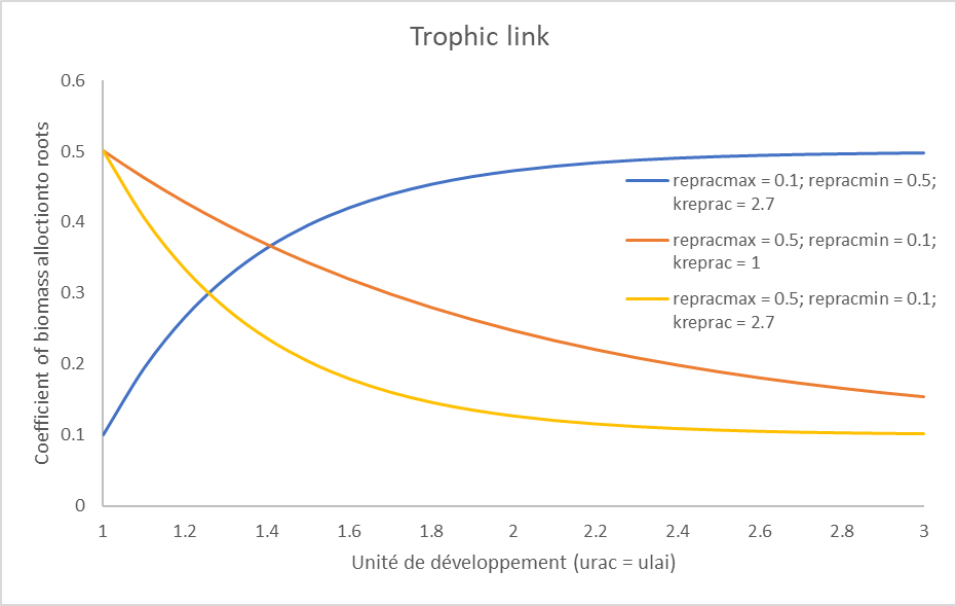
- Users must or should activate different options to have an accurate simulation of the root system :

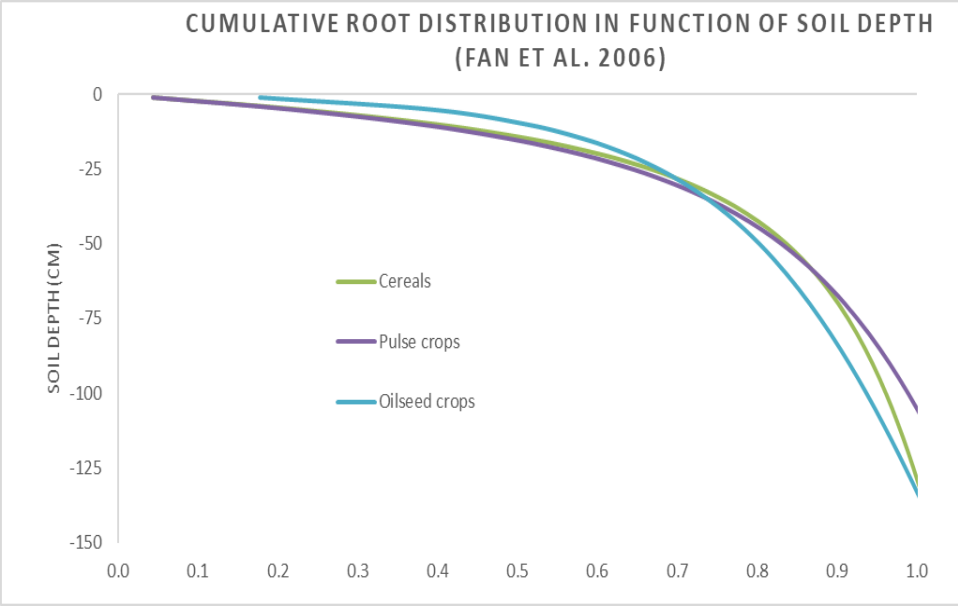
Mandatory options		Definition
coderacine	2	Simulation of true root density and associated biomass
codtrophrac	1	Simulation of a permanent trophic link between daily crop biomass production and daily root length growth
code_rootdeposition	1	Simulation of root N demand and daily C&N fluxes to the soil due to root turn-over
codedisrac	1	Simulation of a standard root distribution within the soil profile

Facultative options		Definition
code_diff_roots	1	Simulation of 2 roots classes: coarse and fine roots with variable specific root length and lifespan
code_stress_root	1	Simulation of a preferential allocation of biomass to roots in case of water on N stress

Mandatory options (exemple: miscanthus)

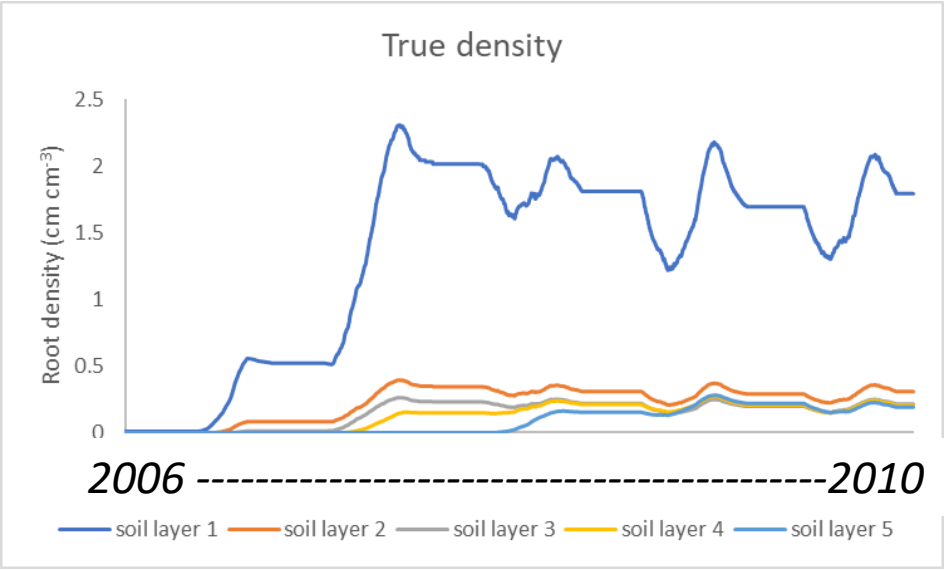
True density option with activation of trophic link, standard distribution and root deposition:

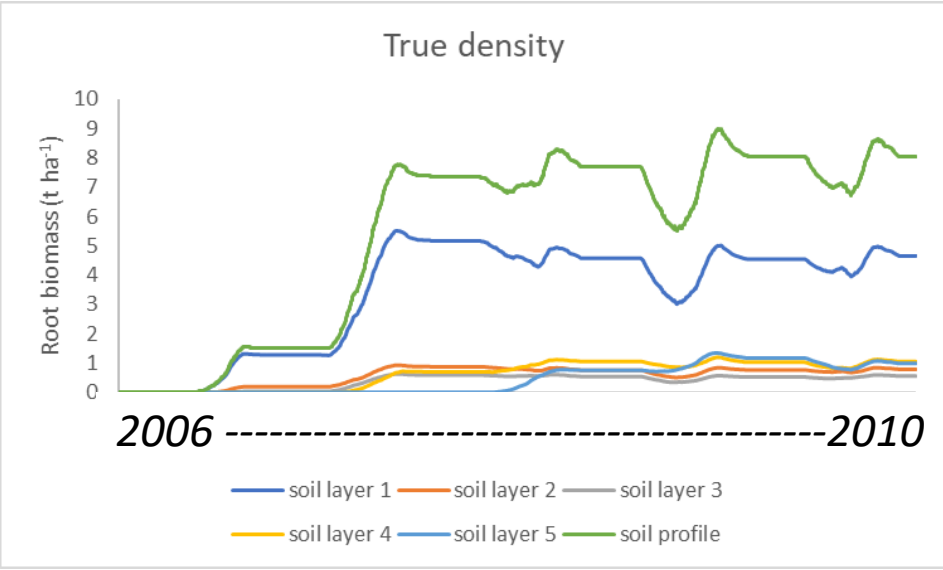


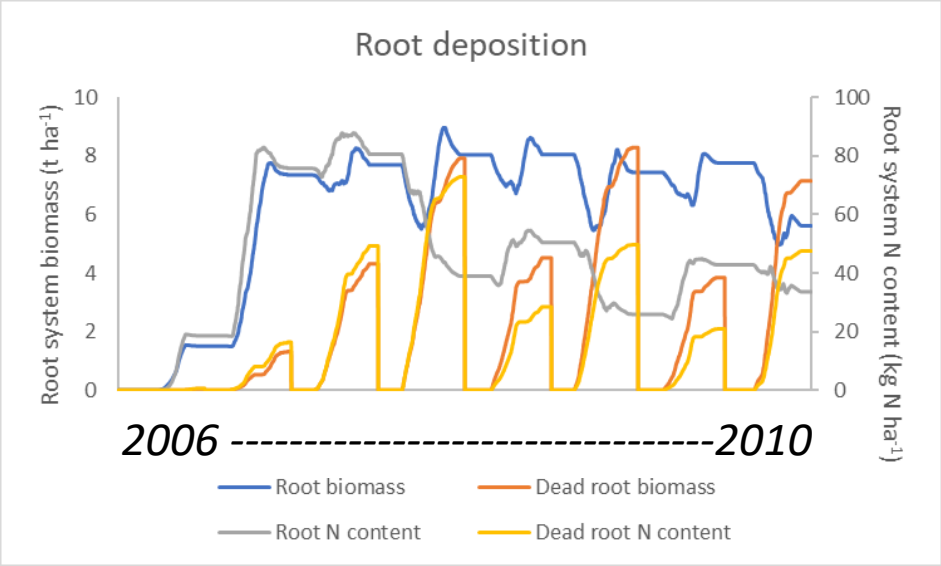


Mandatory options (exemple: miscanthus)

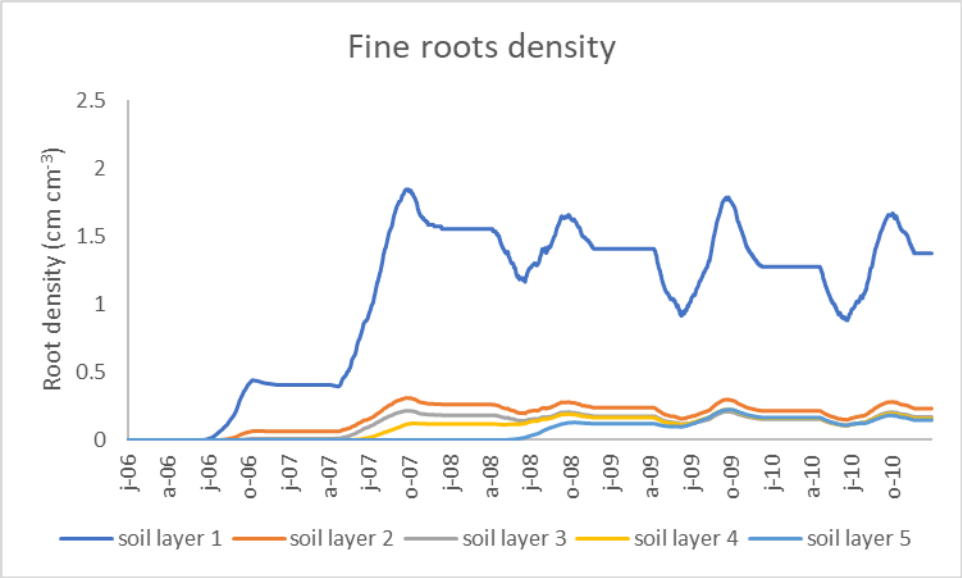
True density option with activation of trophic link, standard distribution and root deposition:

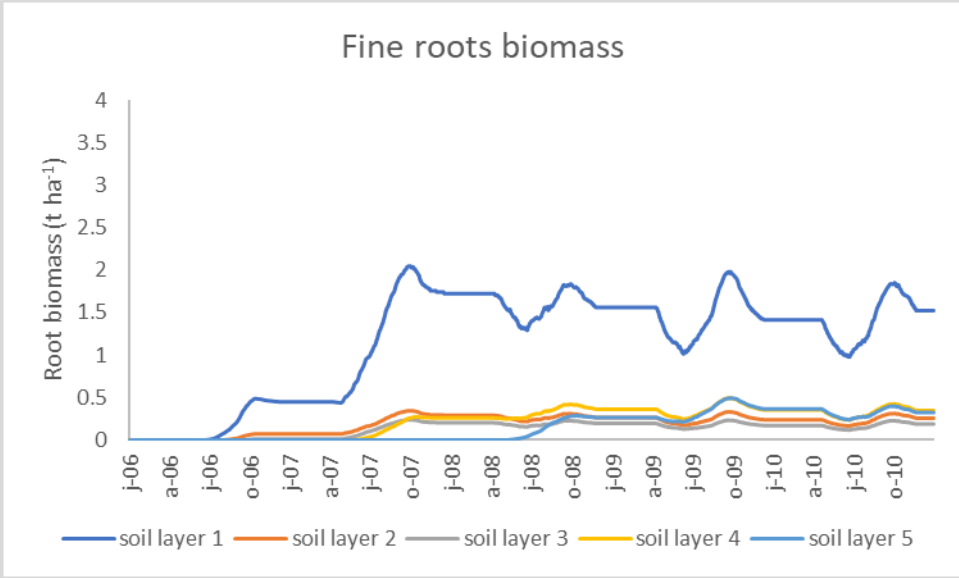


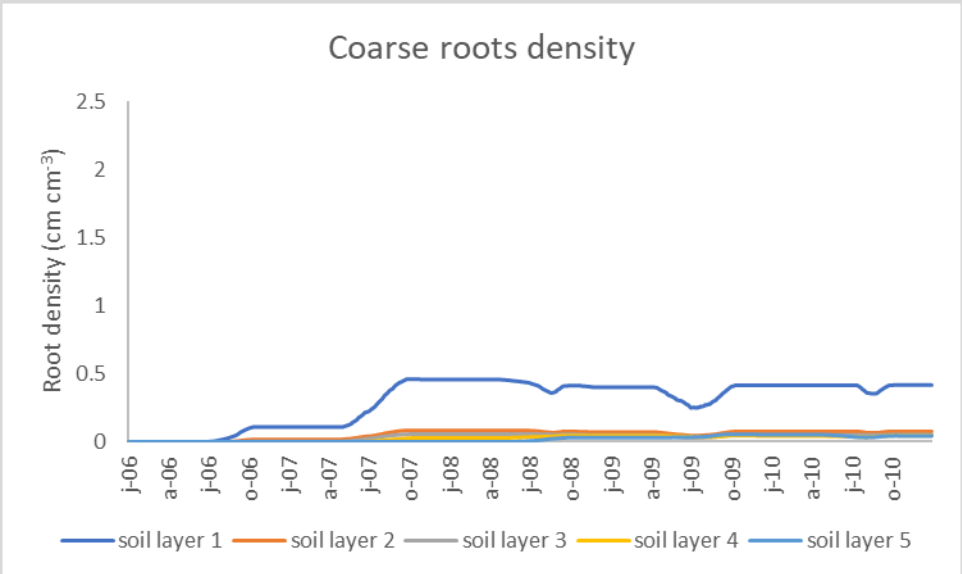


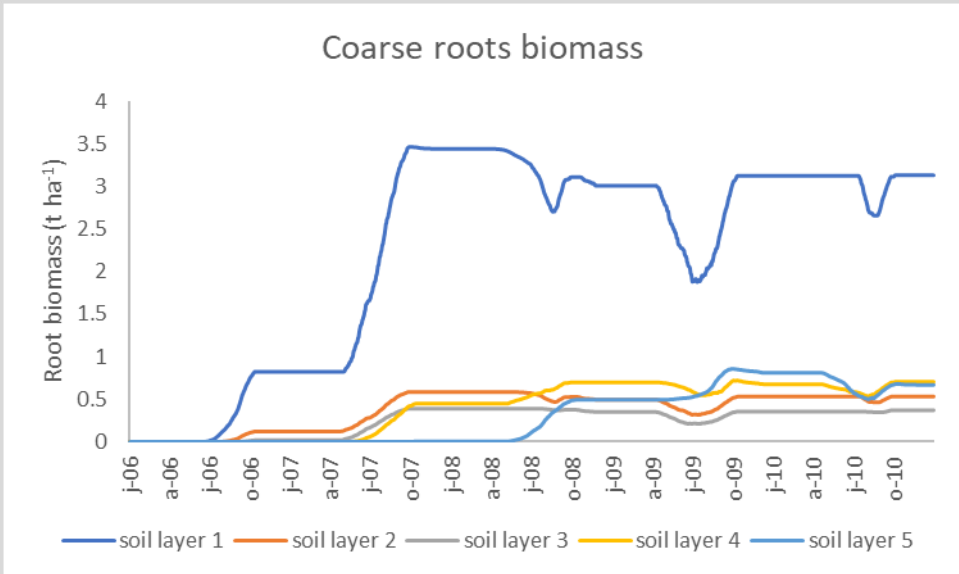


Facultative options (exemple: miscanthus)





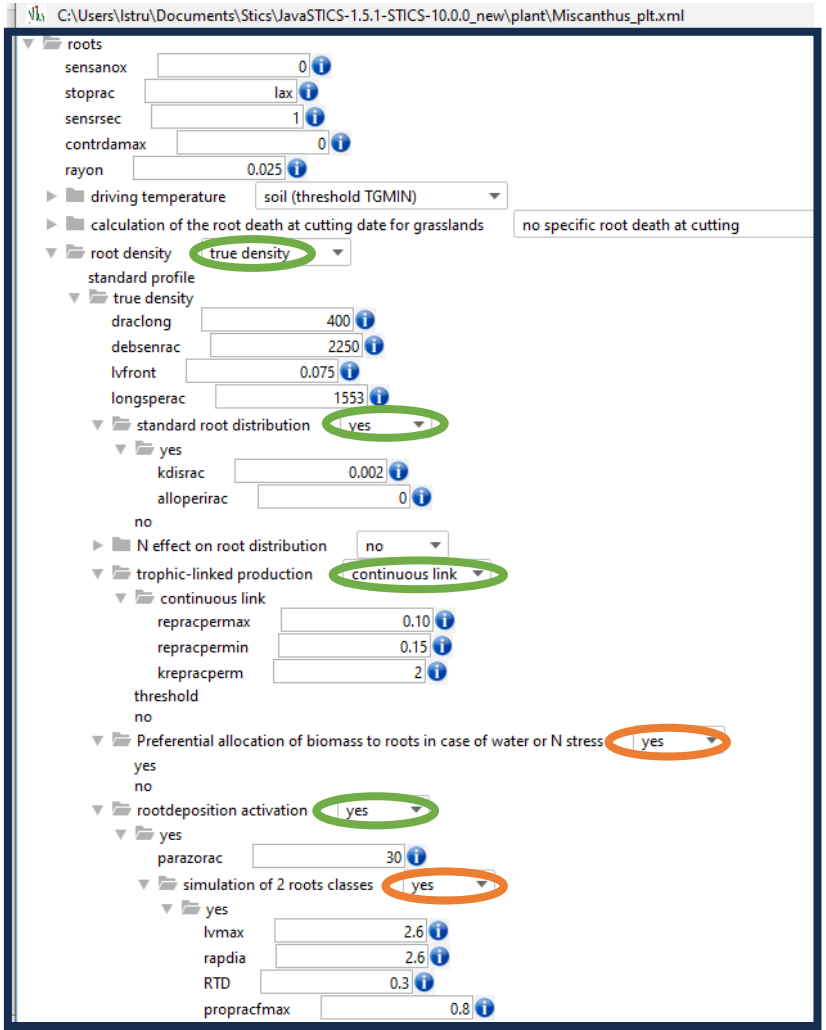




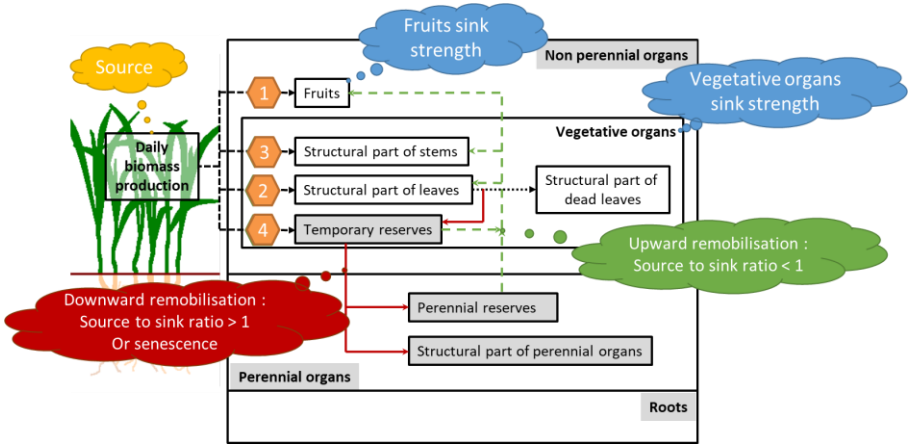
NEW

Root system

- Plant parameters and options for the simulation of root system are available in the plant file:



C& N cycling

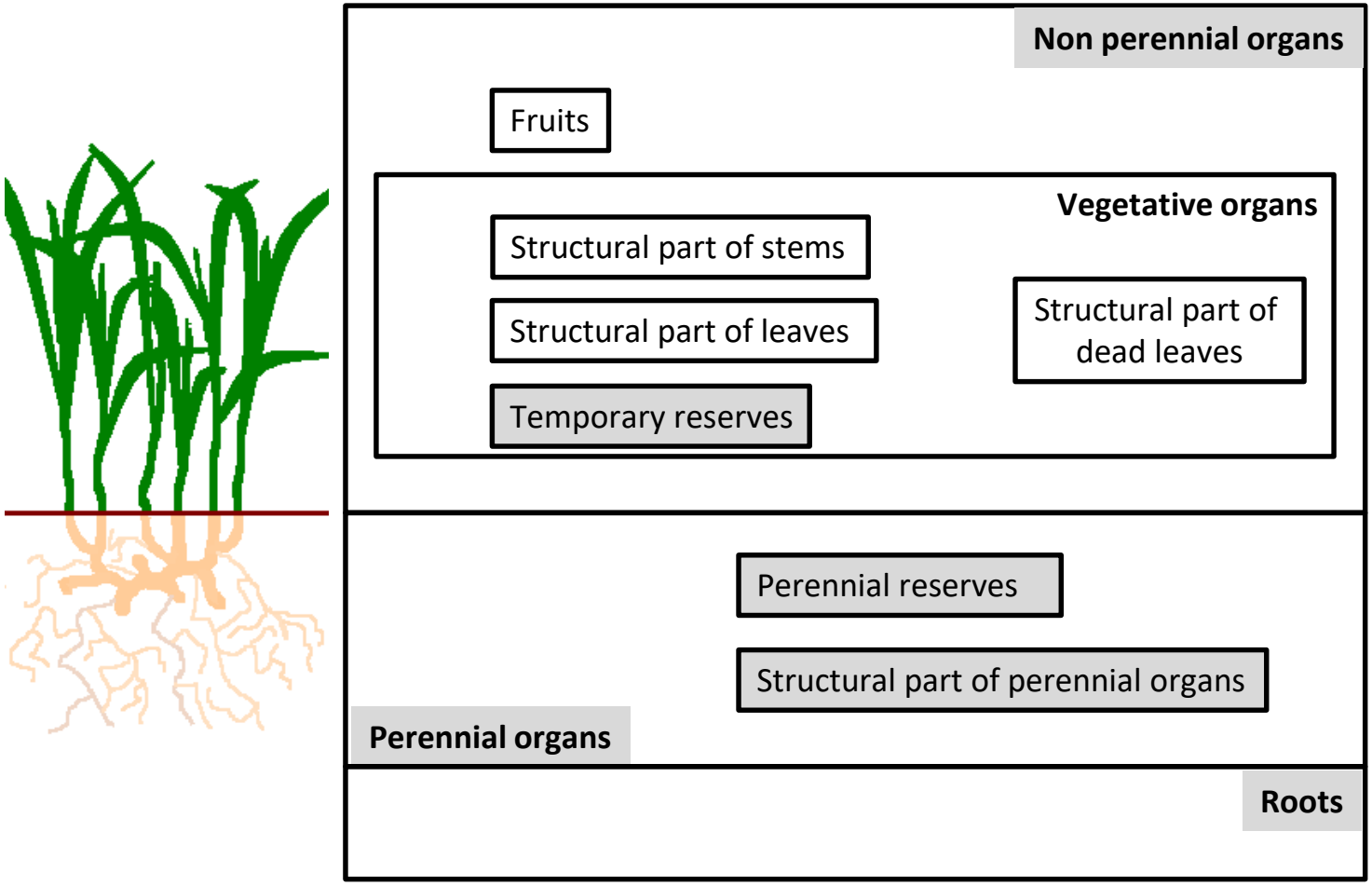


Partitionning : the way the crop is divided into compartments between which the model simulates either carbon (or biomass) fluxes or nitrogen fluxes

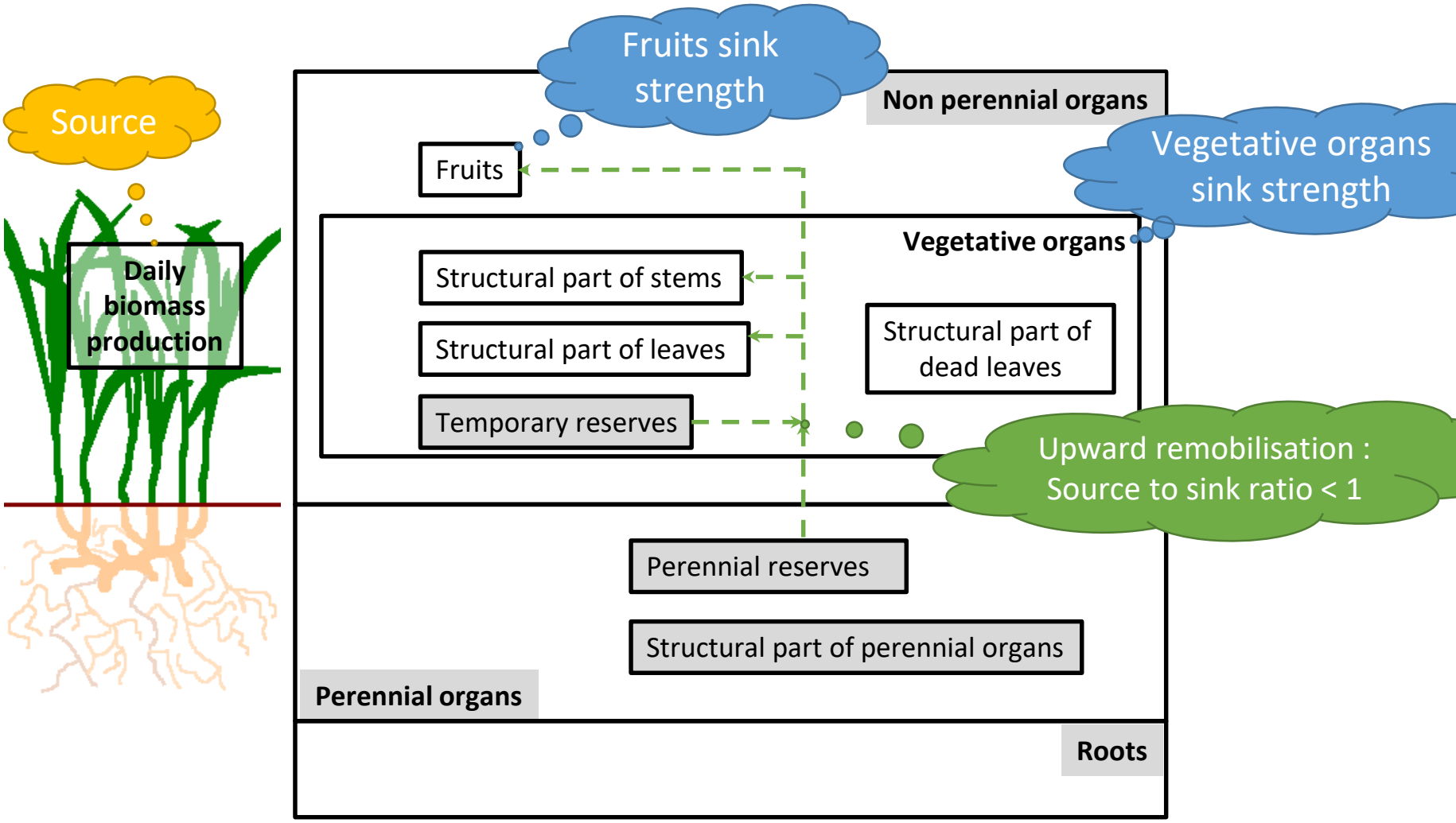
Fluxes:

- Upward remobilisation: remobilisation of C&N from reserves to sustain non-perenial organs growth
- Downward remobilisation: remobilisation to fulfill reserves
- Fluxes to the soil

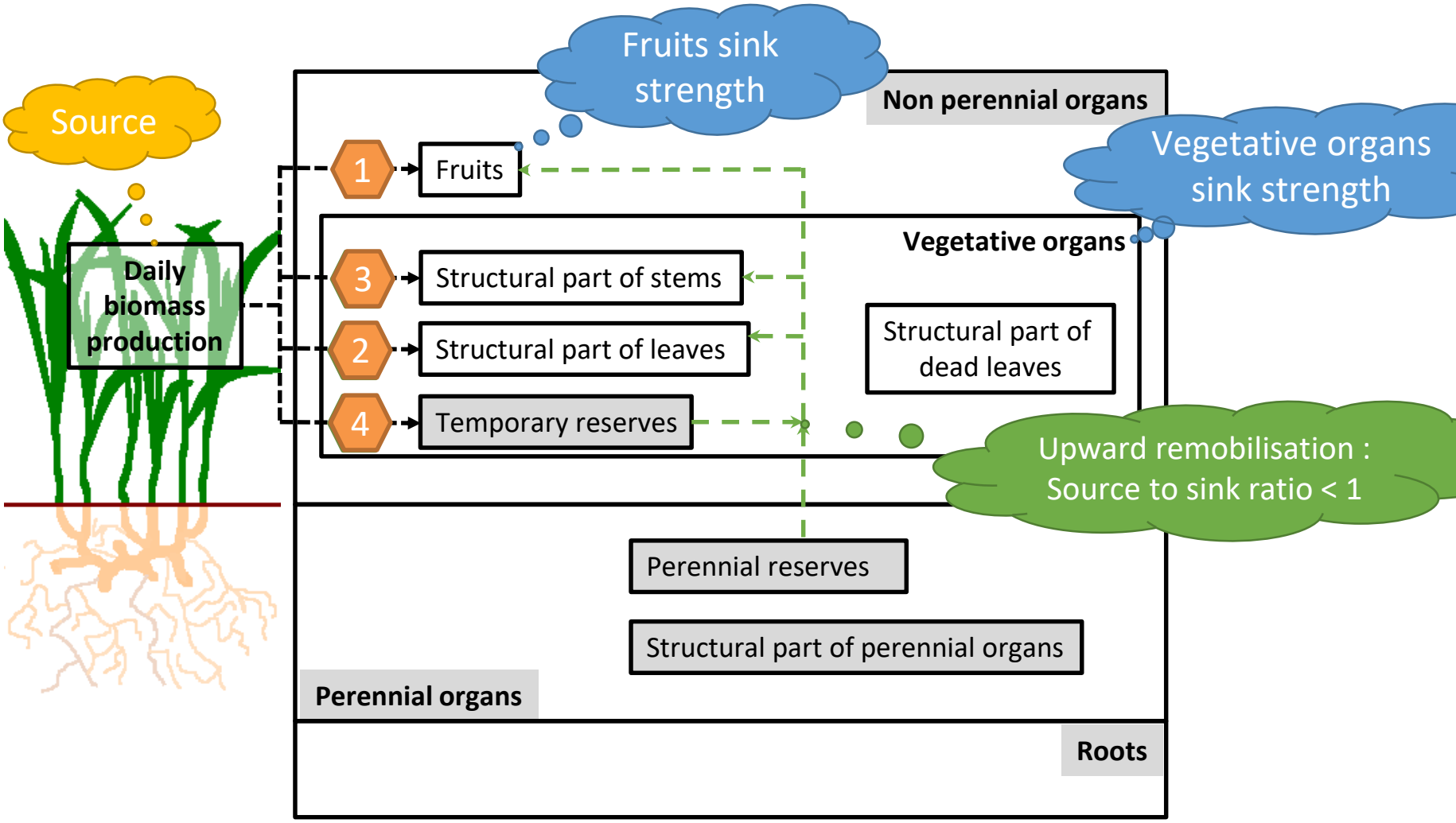
Biomass partitioning in perennial crops (v10) (Chapter 7 pp 131-148)



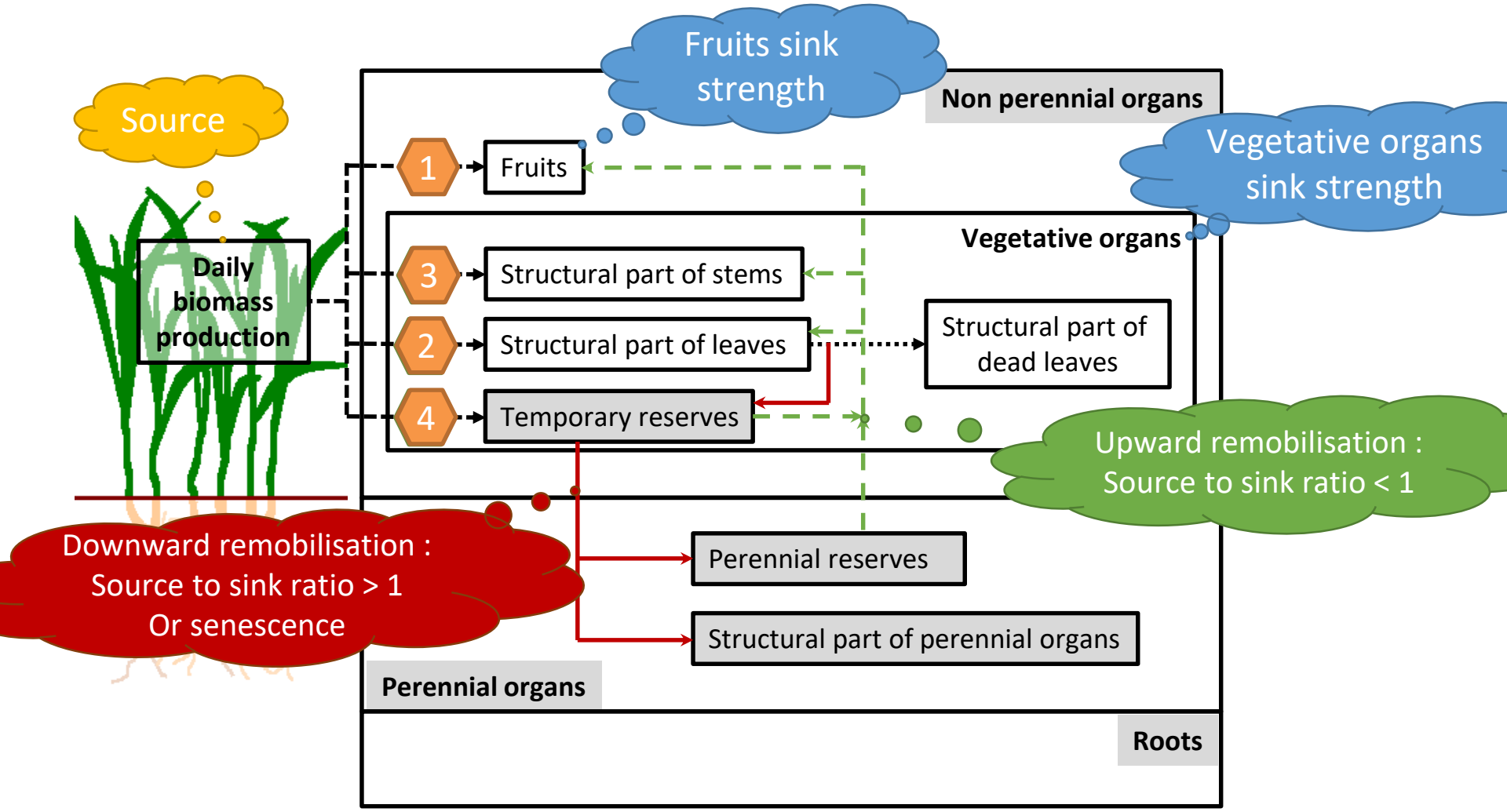
Biomass fluxes in perennial crops (v10)



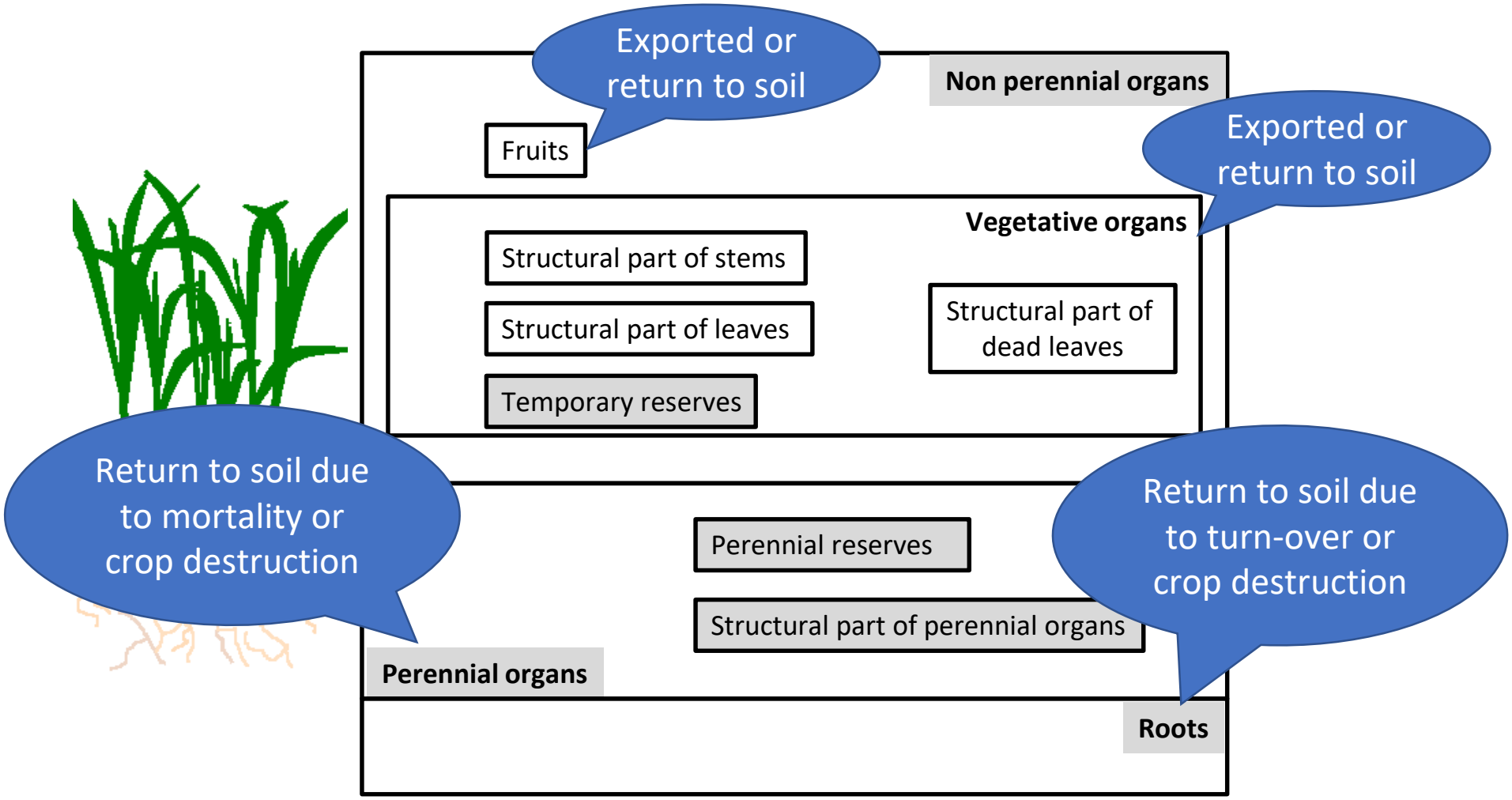
Biomass fluxes in perennial crops (v10)



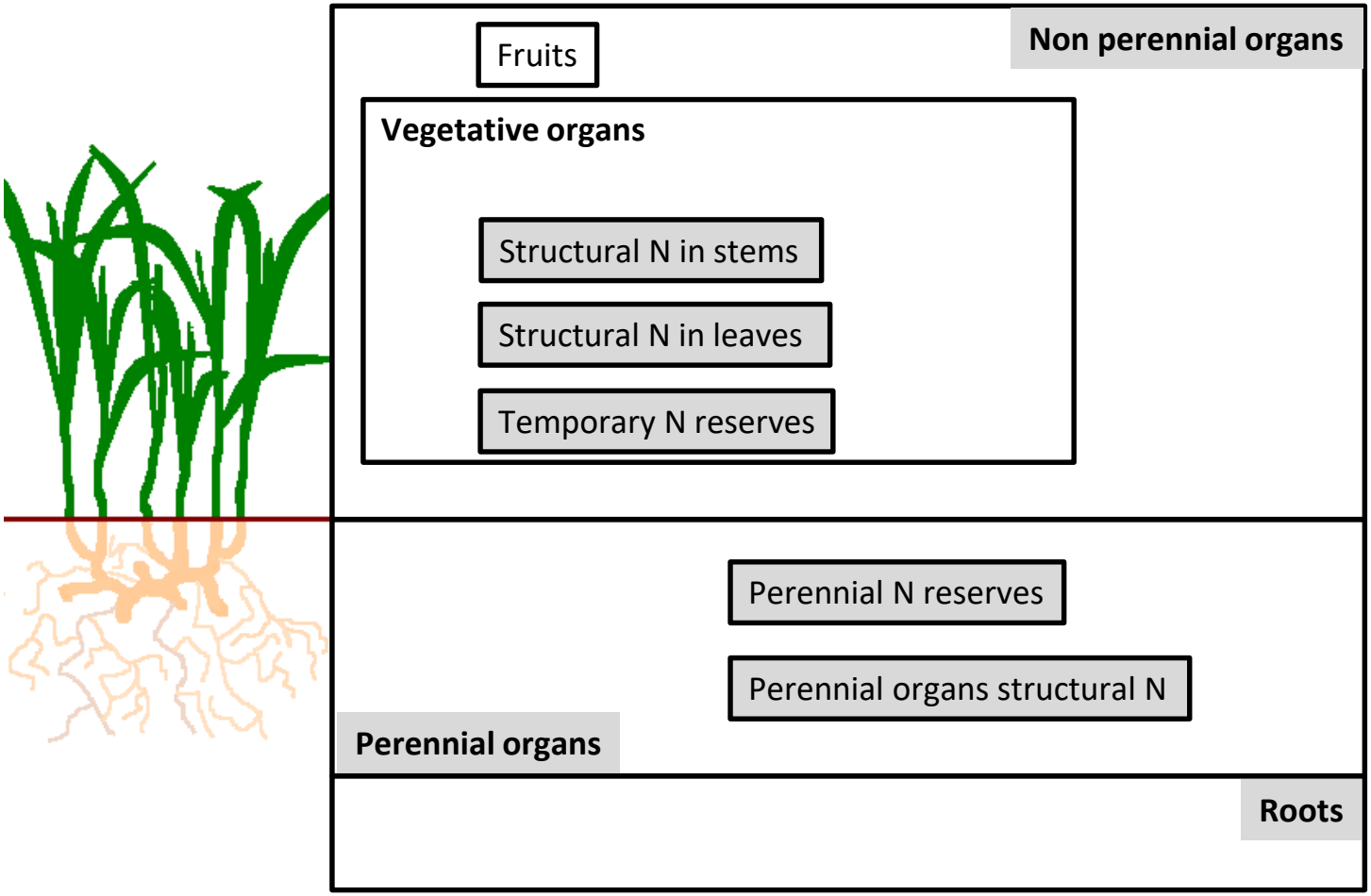
Biomass fluxes in perennial crops (v10)



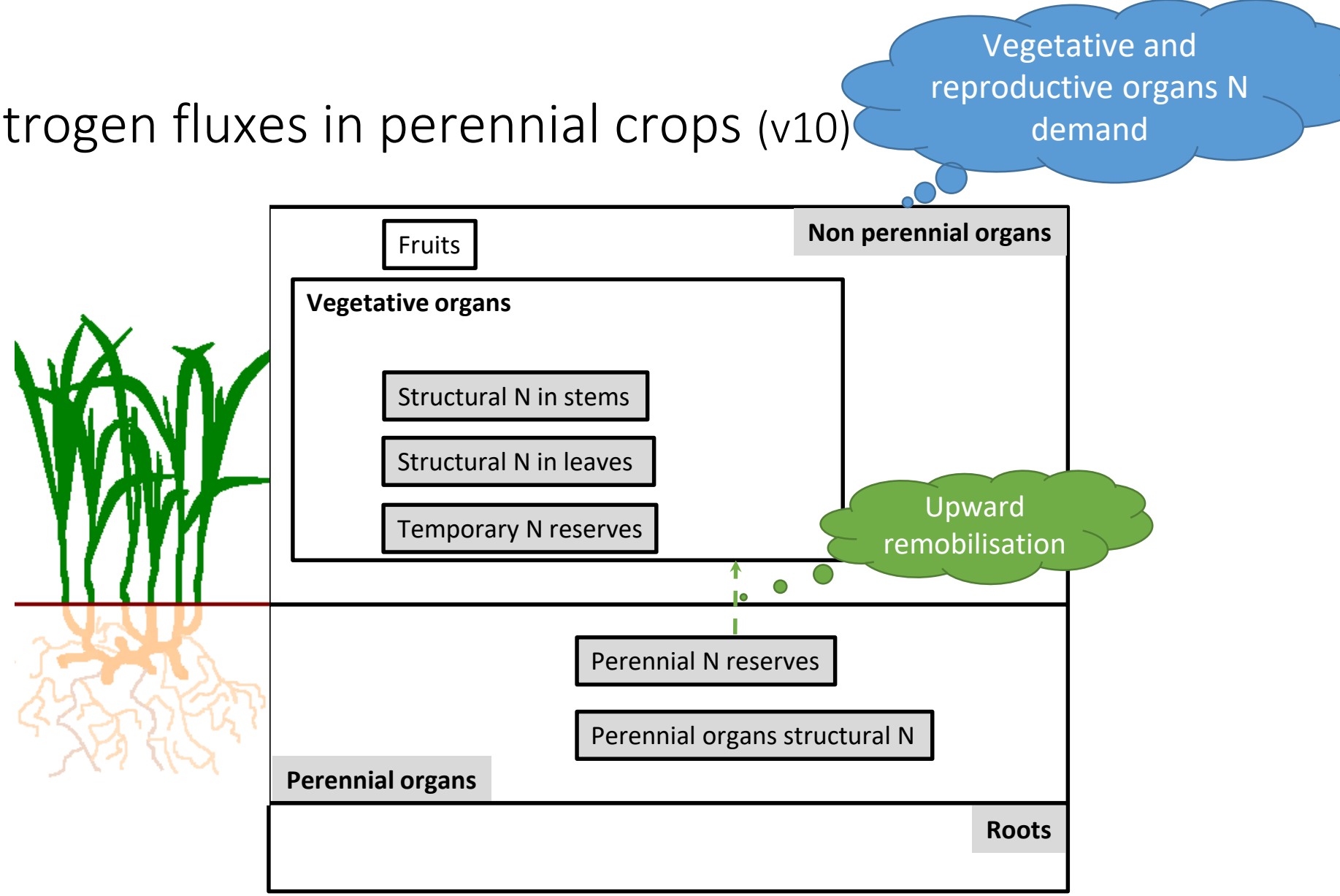
Biomass fluxes to the soil (v10)



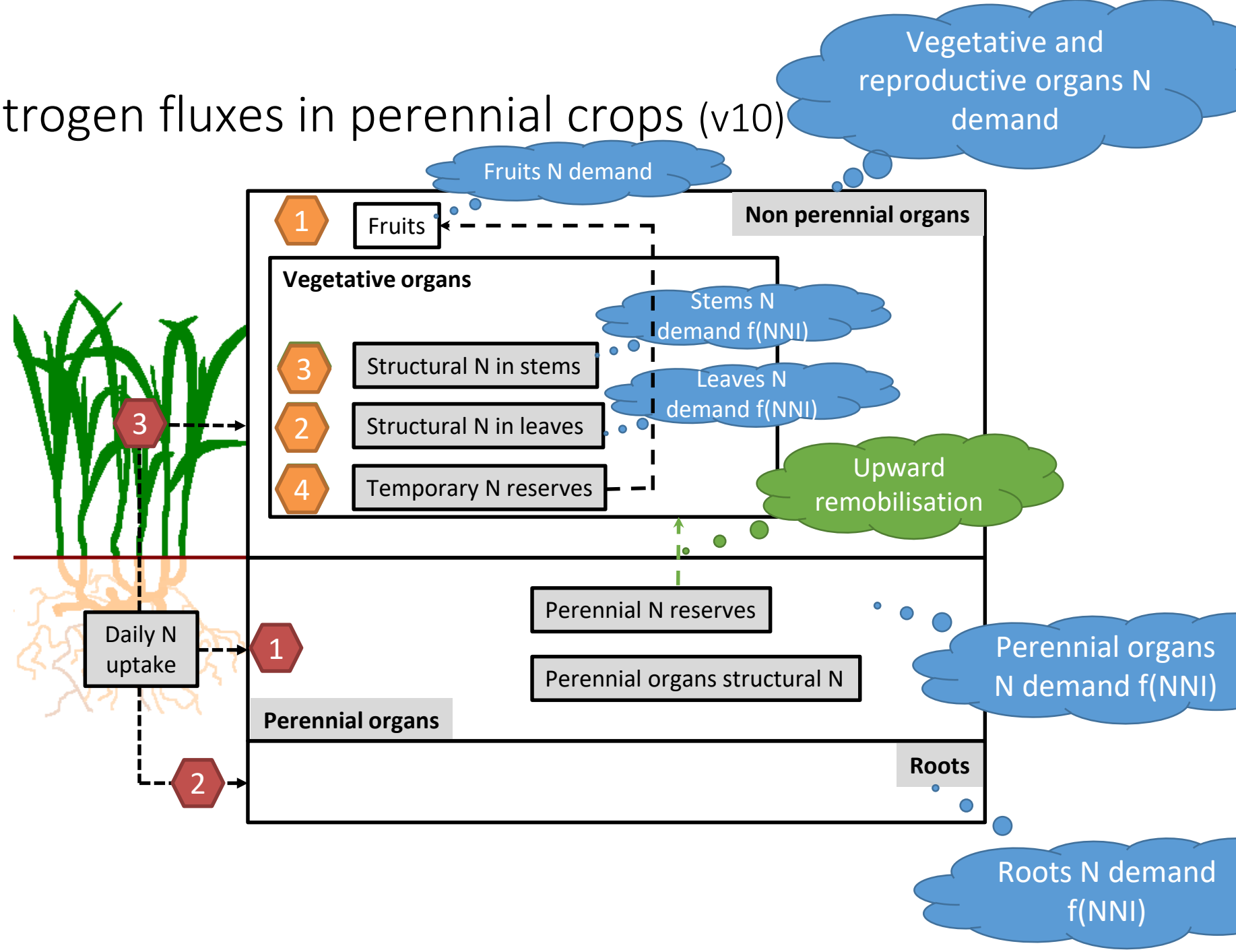
Nitrogen partitioning in perennial crops (v10) (Chapter 7 pp 131-148)



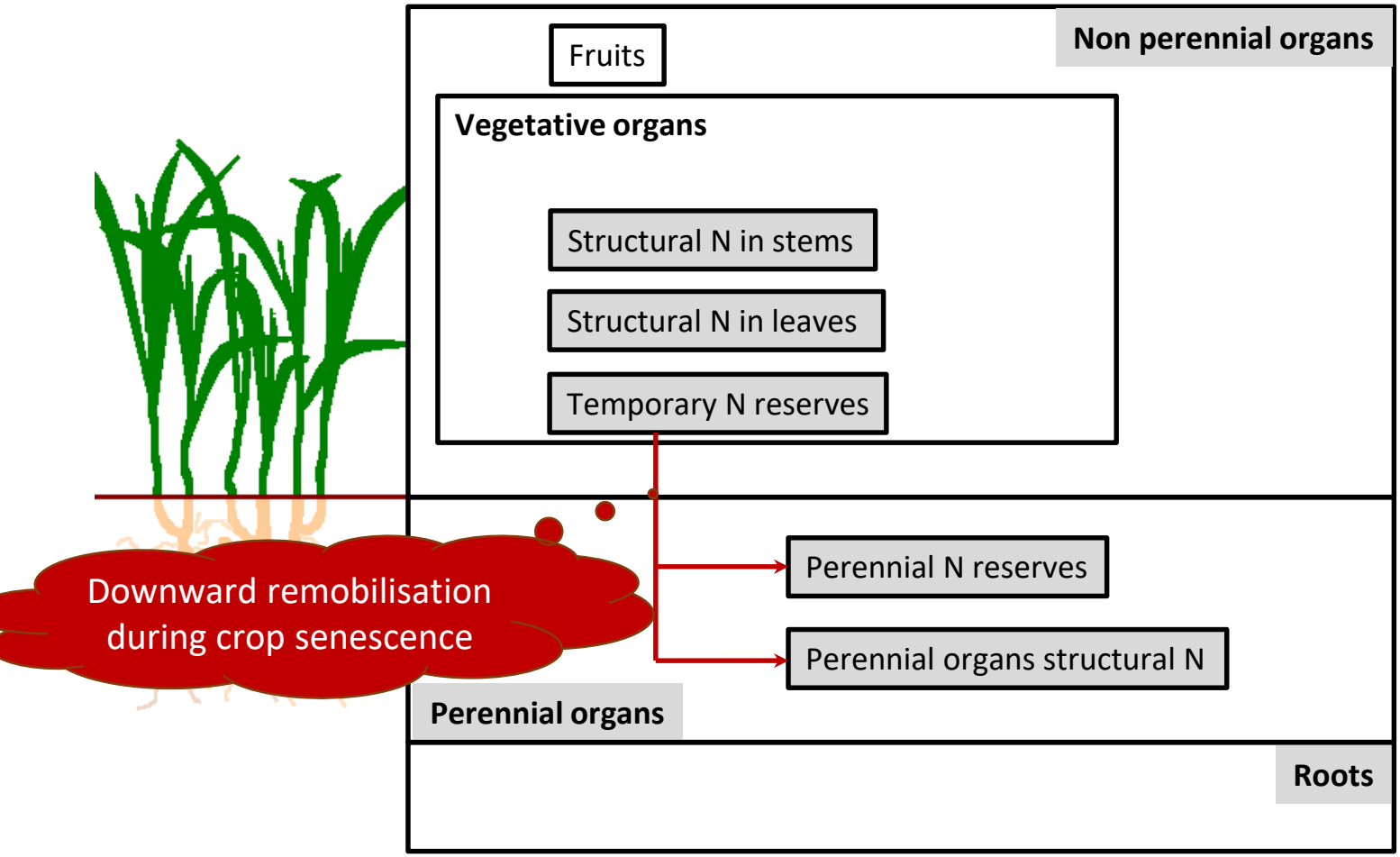
Nitrogen fluxes in perennial crops (v10)



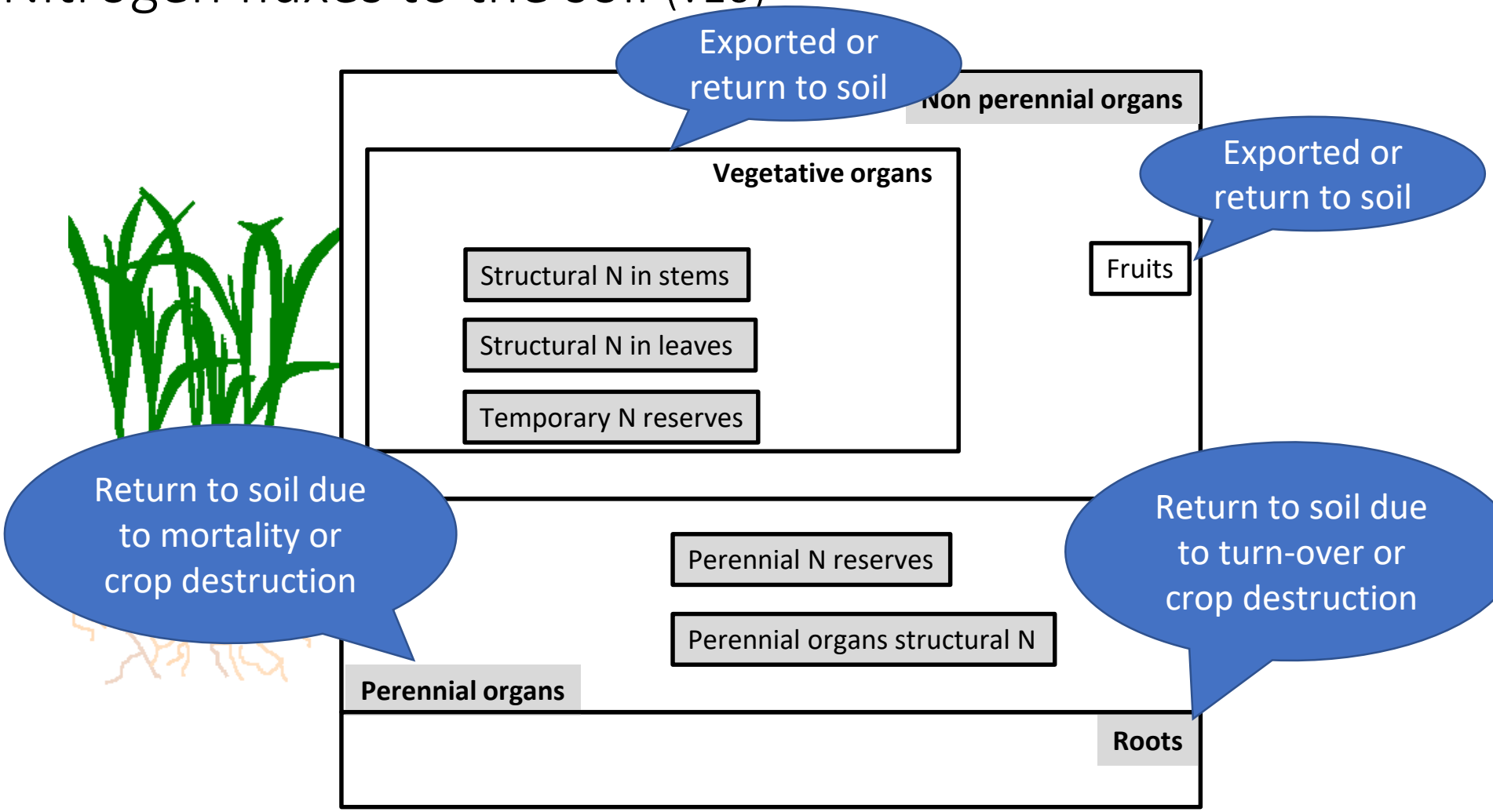
Nitrogen fluxes in perennial crops (v10)



Nitrogen fluxes in perennial crops (v10)



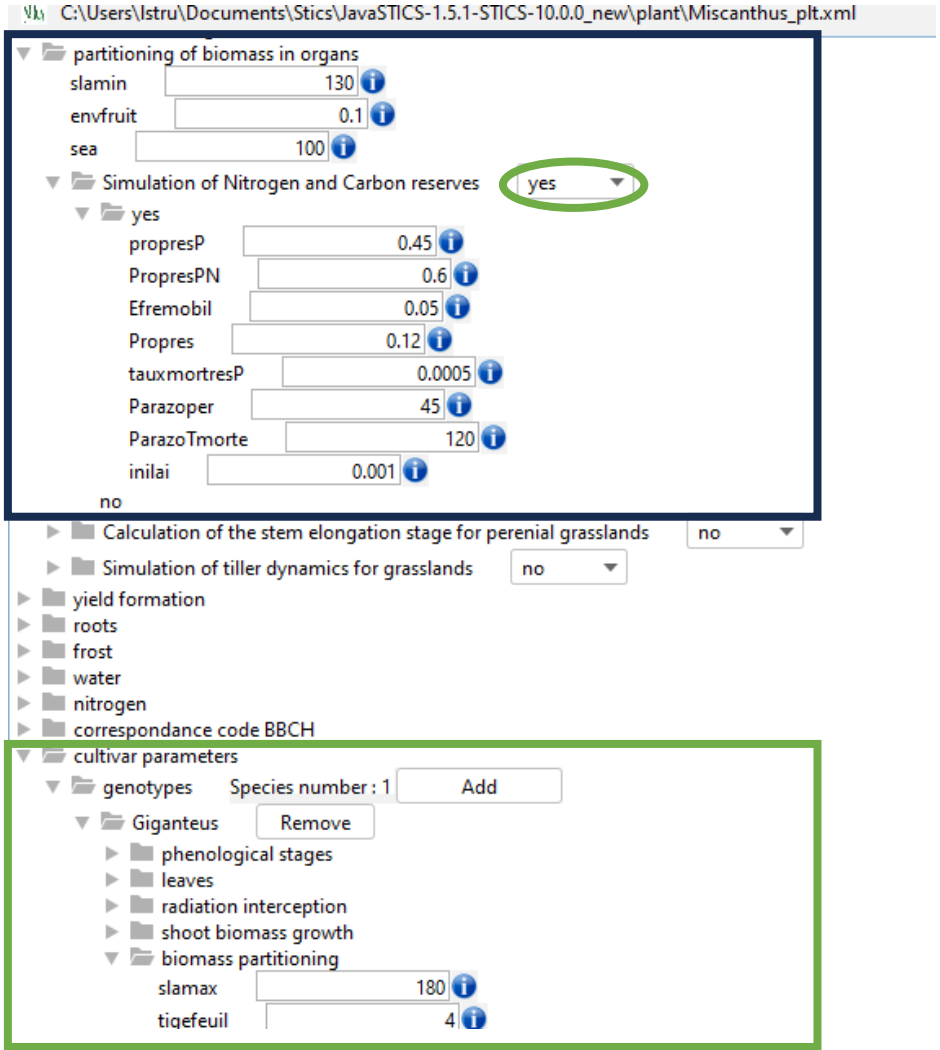
Nitrogen fluxes to the soil (v10)



NEW

Biomass and nitrogen partitioning

- Plant parameters and options for the simulation of biomass and nitrogen partitioning are available in the plant file:



Long term simulation

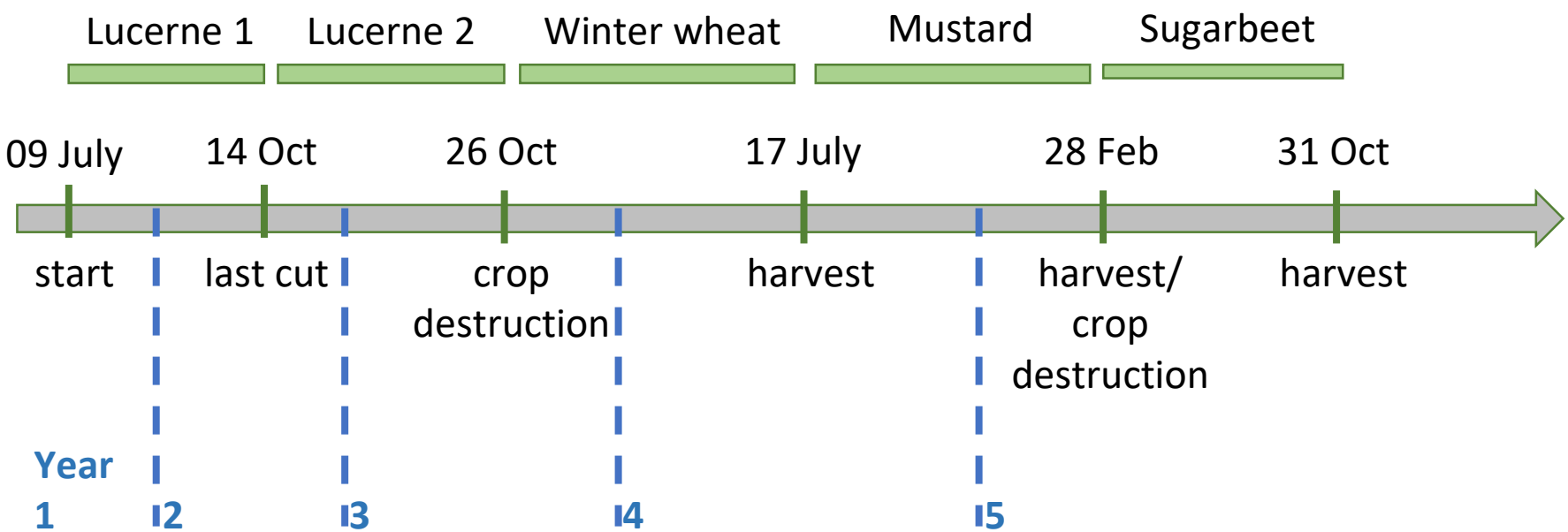
How manage simulations of crop rotations and perennial crops with the STICS model ?

Chapter 14 pp 281-321

How the model enchains the simulations ?

- Model writes state variables at the end of the simulation:
  - Soil variables:
    - *Residues and mulch*
    - *Organic C and N*
    - *Water (by horizon!)*
    - *Mineral nitrogen (by horizon!)*
  - Plant variables for perennial crops or grassland:
    - *Phenology*
    - *Non perennial organs biomass and its partitionning (stems, leaves, temporary reserves), leaf area index and perennial organs biomass and its partitionning between structure and reserves*
    - *Non perennial organs N content and its partitionning (stems, leaves, temporary reserves) and perennial organs N content and its partitionning between structure and reserves*
    - *Root system depth and distribution within the soil profile with corresponding biomass and N content (per centimeter!)*
- Model reads state variables transmitted from the previous USM to initialize the following simulation

How simulate a crop rotation ?



How simulate a crop rotation ?

Lucerne 1 year 1&2	Lucerne 2 year 3	Winter wheat	Mustard	Sugarbeet

...

Lucerne 1 year 1&2	Start of the simulation in summer of year 1 (jj 191) End of simulation = 654 (365 + 289) **/!** 1 day after last cut in autumn
Lucerne 2 year 3	Start of simulation = jj 290 **/!** 1 day after the end of the previous simulation End of simulation = jj 665 **/!** at least 1 day after crop destruction (jj 299)
Winter wheat	Start of simulation = jj 301 **/!** 1 day after the end of the previous simulation End of simulation = jj 565 **/!** at least 1 day after harvest and before sowing successive crop
Mustard	Start of simulation = jj 201 **/!** 1 day after the end of the previous simulation End of simulation = jj 425 **/!** at least 1 day after harvest and before sowing successive crop
Sugarbeet	Start of simulation = jj 61 **/!** 1 day after the end of the previous simulation End of simulation = jj 305 **/!** at least 1 day after harvest and before sowing successive crop

How simulate perennial crop ?

Miscanthus Year 1	Miscanthus Year 2

...

Miscanthus Year 1	Start of the simulation = jj 1 **/!\ use the dedicated plant file for first year** End of simulation = jj 365 **/!** 366 if leap year
Miscanthus Year 2	Start of the simulation = jj 1 End of simulation = jj 365 **/!** 366 if leap year

How to define crop residues returning to soil ?

- Simulation of crop residues is set for all species in the technical files :

  - *Ressuite* parameter in « harvest » = pruning (grapevine), roots (alfalfa, miscanthus, grass…), straw + roots (wheat, pea, barley…), stubble + roots (forage maize…) or whole_crop (catch crop)
    - Option available : code_autoressuite to define the residues as a fraction of unharvested biomass to vegetative biomass at harvest

  - For forage, in « special techniques », « cut crop » with the parameter *tauxexportfauche* (0 = all harvestable biomass return to soil; 1 = all harvestable biomass is exported).

  - For grapevine, with an automatic calculation of residues C&N returning to soil due to topping or pruning (in a next version of the model…)

  - For the destruction of a perennial crop, in « special techniques », « date of plant destruction » with the parameter *jourdes* (to use only for perennial crop, annual crops die automatically after harvest).

  Good to know : root system residues returning to the soil are always simulated but in different ways (function of options used for root system simulation)

How to define crop residues returning to soil ?

C:\Users\lstru\Documents\Stics\JavaSTICS-1.5.1-STICS-10.0.0_new\Practical_Work_2_teachers_done\Wheat_2011_tec.xml

- harvest
  - irecbutoir: 731
  - ressuite: stubble+roots
  - automatic calculation of crop aerial residues in function of user parameterization: no
  - method of harvest: cutting
  - harvest decision: physiological maturity
  - Decision of harvest for associated crops: Maturity of the both (2 dates)
  - rules of harvest/moisture status of the soil: no
- special techniques
  - cut crop: yes
    - yes
      - dynamic calculation of residual lai on biomass after cutting: no
      - reference thermal time to compute cutting dates: in upvt
      - Method.of.cutting: calendar in days
        - automatic
        - calendar in days
          - cutting management — Intervention(s) number : 1 — Add

julfauche	hautcoupe	lairesiduel	msresiduel	anitcoupe	engraiscoupe	tauxexportfauche	restit	mscoupemini

        - calendar in degree days
    - no
  - mulch: no
  - topping: yes
    - no
    - yes
      - largrogne: 0.00000
      - hautrogne: 0.00000
      - biorognem: 0.00000
      - topping calendar: fixed date
  - thinning: no
  - leaf removal: no
  - pruning: no
    - no
    - yes
  - trellis system: no
  - greenhouse crop: no
  - date of plant destruction (for perennial crops only): yes
    - yes
      - juldes: 999

GOLD RULES

- About simulations chaining:
  - **Avoid chaining simulations** if you can, **the simplest is the better!** You can simulate until 2 years without chaining, so enjoy!
  - **Chain simulations just after harvest** of the previous crop **or at dates for which water distribution in the soil is the more homogeneous:**
    - For exemple, in Northern France, at the end of February or beginning of March for a spring crop following a catch crop. You have to adapt it to your climatic conditions!
  - The model only transmit average soil water and N content per layers from an USM to another, so **you loose information when chaining USMs!**
- **Always think that the model doesn't do what you think it does!**
  - The **order of modules calling** in the model can have an **impact on simulations!** (for example soil tillage is called before harvest)
  - **Always check** that the model reproduced well the technical practices that you have parameterized!

FIN

lstrullu@gmail.com